

SOP-C101-SAFE-003 · C-101 Reciprocating Compressor Safety & Maintenance

BPCL Mathura Refinery · Rotating Equipment Division

Document Control

Asset: C-101 · Natural Gas Reciprocating Compressor. Revision: 4. Date: 22-Sep-2023.
Plant: BPCL Mathura Refinery, Unit 3 · Gas Recovery. Compliant with: OISD-105,
OISD-117, OSHA 1910.147, API 618.

1. LOTO (Lockout-Tagout) Procedure · OISD-105 Compliant

MANDATORY per OISD-105 Section 5.3 and OSHA 1910.147. Step 1: Obtain Permit to Work (PTW) No. from Area Operations Supervisor. Step 2: Notify all personnel in compressor bay. Step 3: Shutdown compressor via DCS · wait for coast-down (3 minutes minimum). Step 4: Close inlet gas valve XV-101A and outlet valve XV-101B · lockout with padlock. Step 5: De-energize MCC-3 breaker CB-301 · lockout with personal padlock. Step 6: Tagout all isolation points with Danger Tags. Step 7: Vent and purge with nitrogen · 3 purge cycles minimum. Step 8: Gas test · verify 0% LEL before entry. Document in PTW.

2. Electrostatic Grounding

Per OISD-GDN-180 and OISD-117 Section 7.2: All portable equipment, hoses, and containers must be bonded and grounded before any work involving flammable gas or liquids. Connect grounding cable to designated ground lug GR-C101 (copper lug, painted yellow). Verify ground resistance less than 10 ohms using megohmmeter. Anti-static coveralls mandatory. No synthetic clothing.

3. PPE Requirements · OISD-105 Mandatory

Per OISD-105 Section 4.1 and OSHA 1910.132: Minimum PPE for all compressor maintenance: Full face shield + safety glasses. Chemical-resistant gloves (nitrile 8mil). Flame-resistant coveralls (NFPA 70E Category 2 minimum). Steel-toed safety boots. Hard hat (Class E, electrical rated). Hearing protection (earmuffs, NRR 28 minimum). H2S detector (personal) · mandatory for gas service.

4. Confined Space Gas Testing and Ventilation

Per OSHA 1910.146 and OISD-105 Section 8.2: Compressor crankcase is classified as a Permit-Required Confined Space. Atmospheric testing required before entry: O2 19.5-23.5%, LEL less than 10%, H2S less than 1 ppm (BPCL standard, stricter than OSHA 10 ppm TWA). Continuous forced ventilation (minimum 4 air changes/hour) required during work. Station attendant must be present outside confined space at all times. Rescue equipment (SCBA, rescue harness) to be staged at entry point.

5. Tagout Warning Flag Registration

Per OSHA 1910.147 and BPCL PTW System Procedure PTW-BPCL-001: All lockout/tagout points must be registered in the PTW system before work begins. Tagout flags must be red with white text: DANGER · DO NOT OPERATE. Each tag must show: technician name, date, equipment tag, permit number. PTW Control Room copy must be maintained until all tags are removed. No tag removal without PTW closure and supervisor sign-off.